

### Homework 3 - Traveling Salesman Problem

#### Dataset:

TSP51\_dataset.txt with 51 cities

Forbidden edges: [[1, 22], [1, 32]]

#### Design choices:

- Chromosome: permutation of 50 cities with city 1 fixed as start/end
- Fitness:  $1 / (1 + \text{total route distance})$
- Selection: tournament selection
- Crossover: order crossover (OX)
- Mutation: swap / insert / inversion
- Additional solver for comparison: Simulated Annealing

#### Best results:

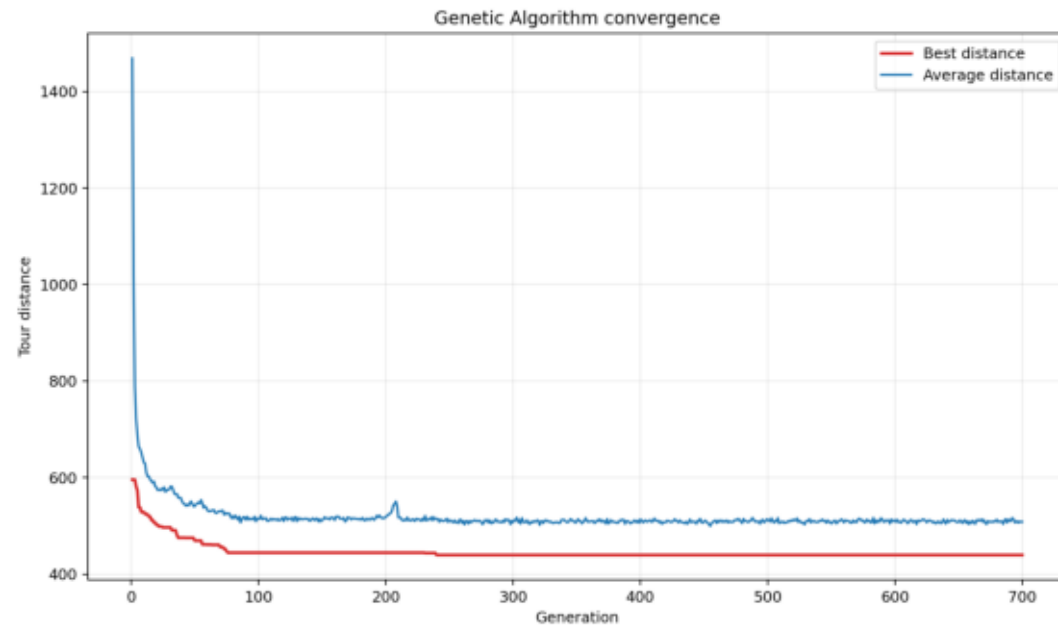
- GA best distance: 439.53 (runtime 21.54s)
- SA best distance: 444.23 (runtime 4.99s)

#### Best GA route:

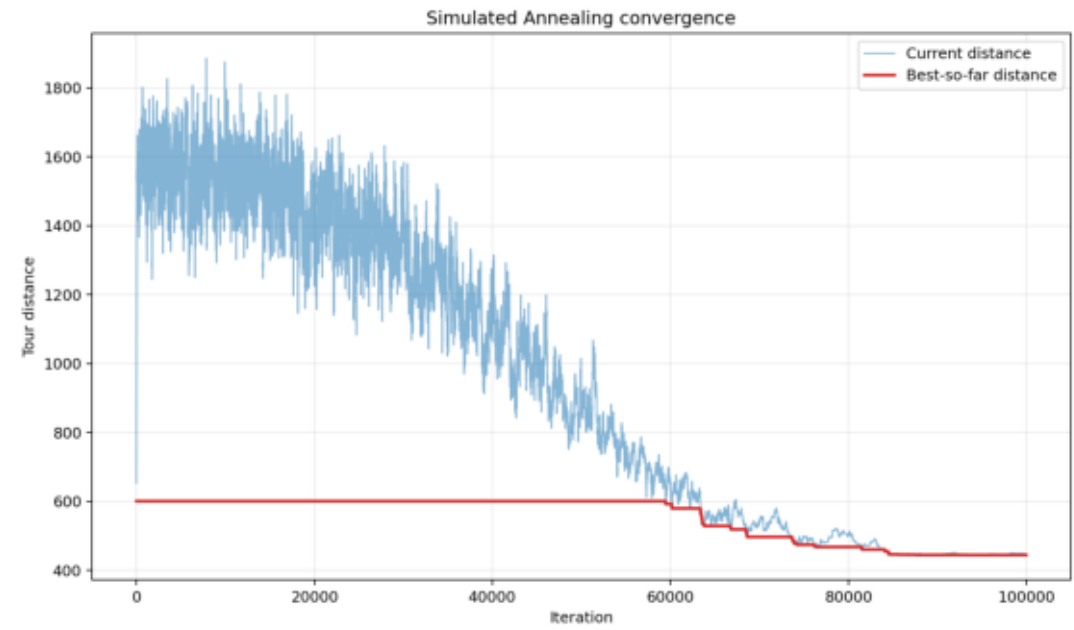
1 -> 8 -> 26 -> 31 -> 28 -> 3 -> 36 -> 35 -> 20 -> 29 -> 21 -> 34 -> 30 -> 9 -> 50  
-> 16 -> 2 -> 22 -> 32 -> 11 -> 38 -> 5 -> 49 -> 10 -> 39 -> 33 -> 45 -> 15 -> 37 ->  
17 -> 44 -> 42 -> 19 -> 40 -> 41 -> 13 -> 25 -> 14 -> 24 -> 43 -> 7 -> 23 -> 48 -> 6  
-> 18 -> 4 -> 47 -> 12 -> 46 -> 51 -> 27 -> 1

# Convergence history

## Ga History

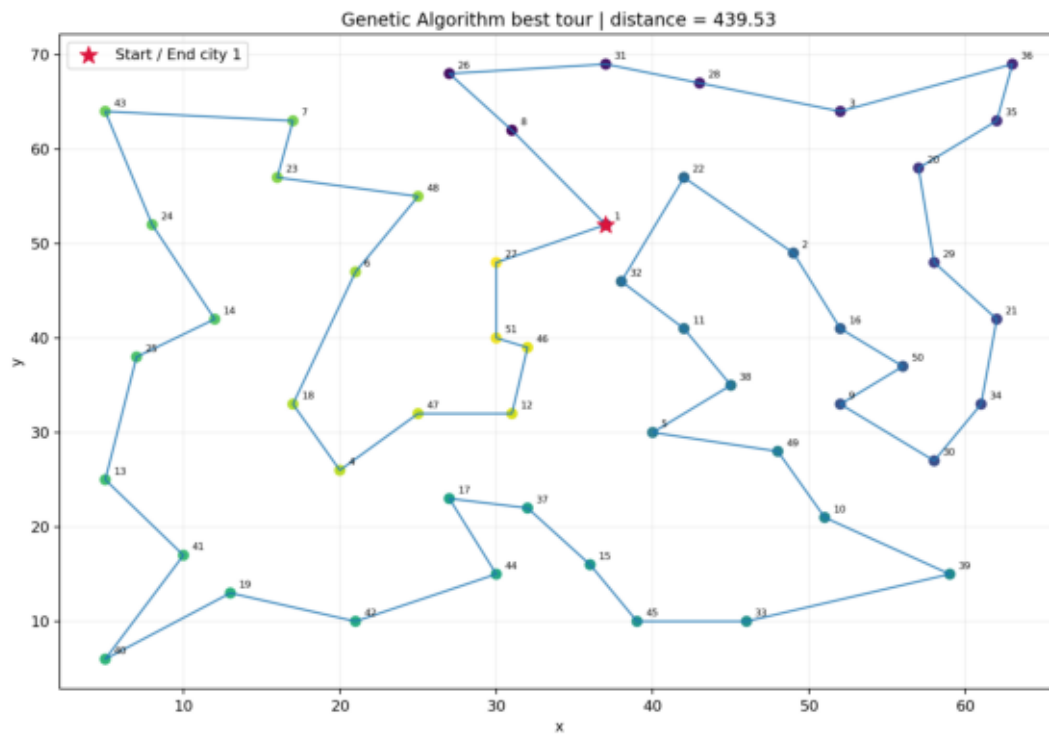


## Sa History

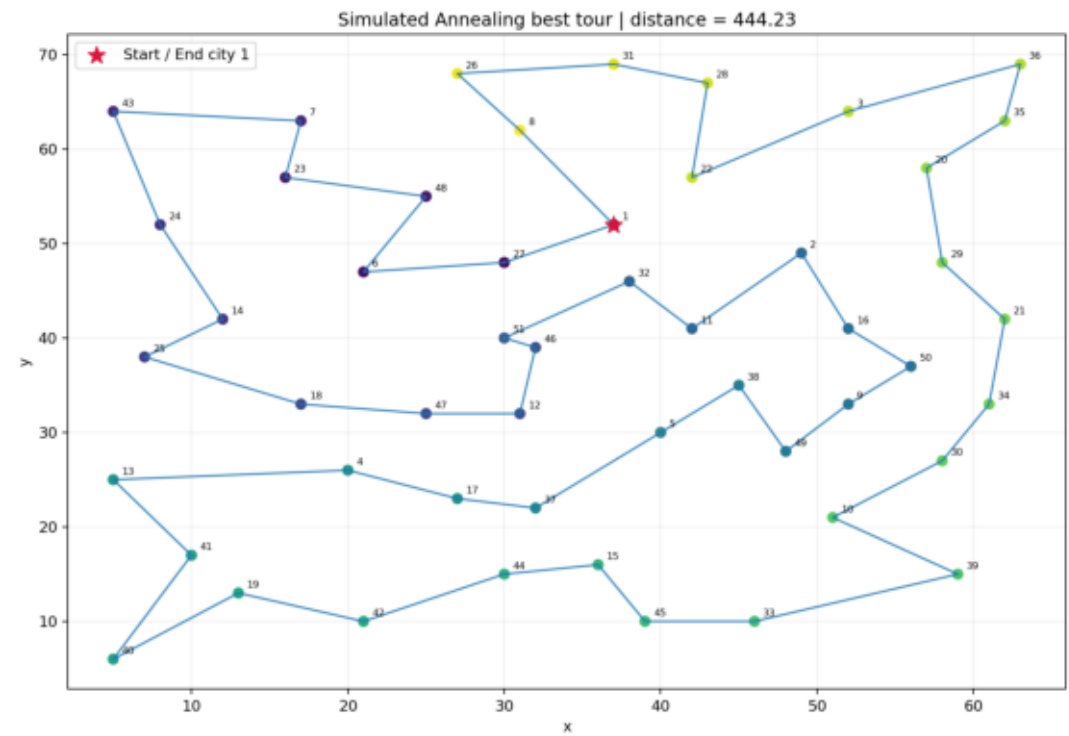


# Best tours

## Ga Route

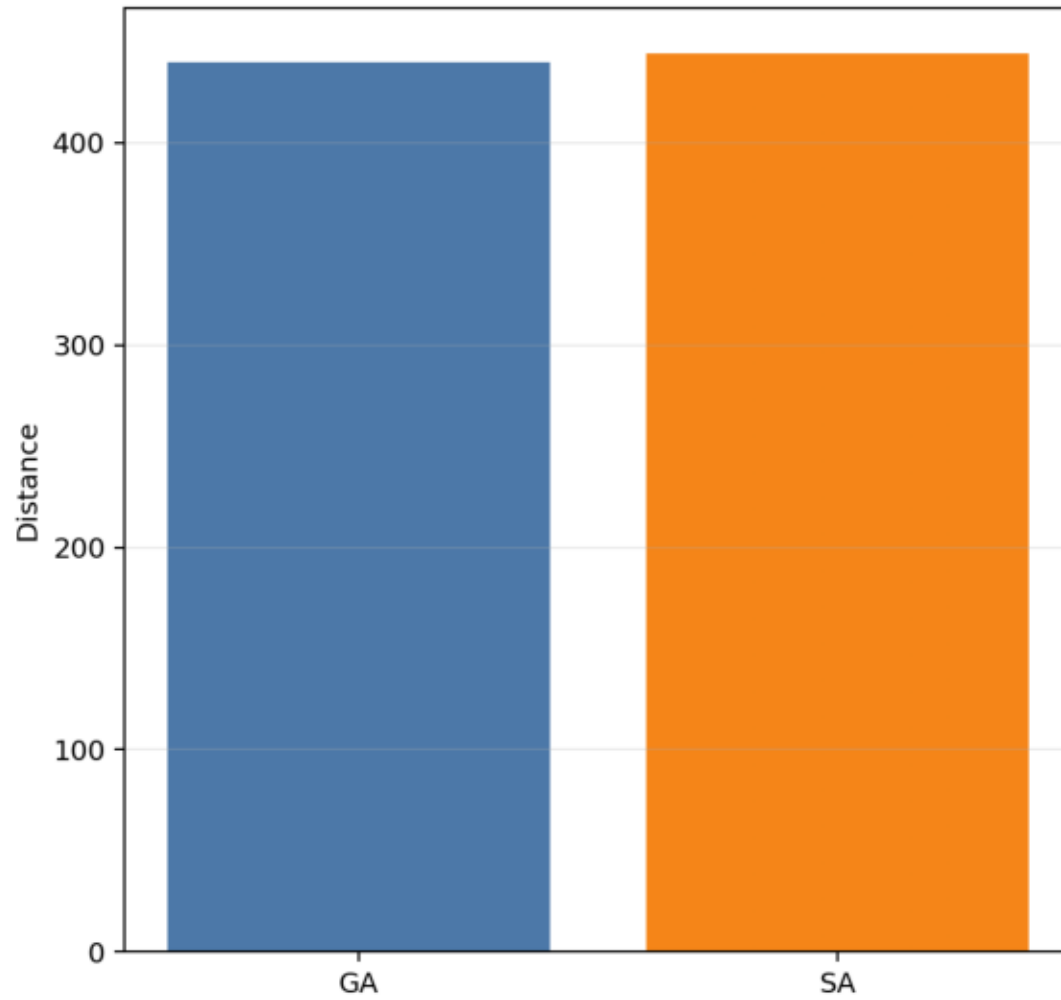


## Sa Route



## Distance and runtime comparison

Best tour distance



Runtime

